



## Converting fractions to recurring decimals

- 1 A decimal with an infinite repeating pattern of digits is called a recurring decimal. Fill in the blank
- 2 Match each decimal given using dot notation to its equivalent decimal. Write the correct letter in each box

|   |                     |
|---|---------------------|
| a | $0.\dot{1}\dot{2}$  |
| b | $0.12\dot{3}$       |
| c | $0.\dot{1}2\dot{3}$ |
| d | $0.\dot{1}$         |
| e | $0.1\dot{2}\dot{3}$ |

|          |                    |
|----------|--------------------|
| <i>b</i> | $0.123333333\dots$ |
| <i>d</i> | $0.111111111\dots$ |
| <i>a</i> | $0.12121212\dots$  |
| <i>c</i> | $0.123123123\dots$ |
| <i>e</i> | $0.123232323\dots$ |

- 3 Select all the fractions which are equivalent to a recurring decimal. Tick **3** correct answers

☒  $\frac{1}{3}$

☒  $\frac{1}{12}$

☐  $\frac{3}{24}$

☐  $\frac{6}{15}$

☒  $\frac{10}{12}$



4 Use short division to write  $\frac{8}{15}$  as a decimal. Tick **1** correct answer

☐ 0.53

☐ 0.5333333333

☒  $0.5\dot{3}$

☐  $0.\dot{5}3$

5 Jacob uses his calculator to write  $\frac{11}{12}$  as a decimal. His calculator display shows the number 0.9166666667. Jacob says, "This shows my fraction terminates." Is Jacob correct? Explain why. Tick **1** correct answer

☐ Yes; Jacob's number has 10 digits after the decimal point so it terminates.

☐ Yes; the last digit is 7 not 6 so it doesn't repeat. So it must terminate.

☒ No; the calculator only has a 12 digit display, the last 6 is rounded up to 7.

☐ No; all decimals recur eventually.

6 Izzy writes the recurring decimal  $0.71\dot{6}$  as the fraction  $\frac{\square}{60}$ . What number should she write in the square? 43 Fill in the blank

